

The Universal Observable Pipeline for Sectorized Observable Frameworks

SOF Realization and Cross-Species Registry Evidence

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This paper is Part X of the RIME program. Paper VIII introduces the Sectorized Observable Framework (SOF) as the static object language [7]. Paper IX introduces observable trajectories over SOFs [5]. Paper X isolates the common observable pipeline through which Rubik, quantum, stochastic-finance, Markov, graph, neural-network, and Yang-like systems become comparable.

Abstract

Problem. Papers VIII and IX define the static and dynamic SOF architecture. The remaining question is why different species can share the same observable architecture at all. The issue is not whether every mathematical system is an SOF, nor whether all registered systems obey the same dynamics. The issue is whether represented or finite observable systems pass through a common realization pipeline before their diagnostics are compared. This is not an extension of Wedderburn–Artin structural decomposition; it is an observable layer built above whatever structural decomposition or native coordinate system the species starts with.

Approach. We formulate the **Universal Observable Pipeline**:

represented or finite species $\longrightarrow$ sectorization $\longrightarrow$ observable family
 $\longrightarrow$ observable shadows $\longrightarrow$ observable diagnostics.

This pipeline is the theorem-level object of Paper X. We then introduce the **SOF Registry** as the evidence architecture for the pipeline. Each registered species must supply a SOF object, observable ladder, dynamics when present, diagnostics, and claim-status metadata. The registry records common diagnostics:

$\tau(O)$, $\text{wall}(O)$, $\text{repair}(O)$, $\text{plateau}(O)$.

Results. The main structural result is the Universal Observable Pipeline Principle: any admitted species equipped with a finite space, a sectorization, and an observable extraction rule induces an SOF and hence a canonical observable ladder. The current registry contains Rubik, naturally occurring Rubik Type III/IV mechanism instances, synthetic Type III/IV boundary controls, Xu-style ridge rate-separation, finite spectral-triple, control, PDE, combinatorial, quantum gate, barrier-option stochastic-finance, Markov, graph, neural-network, and Yang-like filtration instances. These entries are evidence for the pipeline, not the theorem itself. The registry probes include a mechanism-separated SOF control giving an internal H3 positive example, Xu–Vardi–Safran grokking as an independent rate-separation precedent, Yang-like state mixing as a monotone filtration-degeneration contrast with RIME’s oscillatory generator-weight dynamics, and quantum Clifford/CNOT systems as non-Rubik accessibility diagnostics, including entanglement-enabled transport channels.

Implications. Paper X shifts the meaning of universality from “all systems obey the same law” to “all admitted systems pass through the same observable pipeline.” The registry supports, but does not prove by itself, this pipeline view of SOF. Paper X does not claim a universal

wall law or a universal rate theorem. Its main causal claim is the **Calibrated Mechanism-Separation Principle**: when distinct observable channels are driven by distinct mechanisms and the response constants are ordered, observable proxy rates separate for structural reasons rather than by post-hoc fitting. A constructive SOF control realizes this calibrated principle with $\tau(K_0^{\text{grow}}) = 30 \ll \tau(K_1^{\text{decay}}) = 1380$. The broader cross-species universality claim remains registry-level: structured dynamics is the positive condition, while linear interpolation, graph rewiring, and state-mixing probes fail or become degenerate.

Notation Table

Symbol	Meaning
$\mathcal{F} = (V, \{Q_i\}, \mathcal{X})$	Sectorized Observable Framework
$\mathcal{R}$	SOF Registry
O	observable shadow or diagnostic
$\tau(O)$	characteristic time scale of O
$\text{wall}(O)$	locus where O changes qualitative type
$\text{repair}(O)$	frozen-to-accessible transition or analogue
$\text{plateau}(O)$	interval or regime where O remains stable
R_1	direct support / transport shadow
R_2	commutator-survival shadow
D	first-depth accessibility shadow
K_0, K_1, K_2	raw norm proxies for direct, commutator, and nested-depth observables

1 Introduction

Paper VIII answers the static question:

What is the sectorized observable object?

Paper IX answers the dynamic question:

How do observable trajectories evolve over SOFs?

Paper X asks the comparative question:

Which observable phenomena persist across different SOF species?

It also asks the prior structural question:

Why can different species share the same observable pipeline?

This is not a new structural decomposition theorem. Classical Wedderburn–Artin theory decomposes semisimple representations into structural blocks [8, 11, 10]. Paper X starts above that layer and asks how sectorized observables, observable ladders, dynamics, and diagnostics are organized once a structural or native representation has been chosen.

The guiding principle is:

different species, common observable architecture, different deformation geometry, comparable observable diagnostics.

The answer is the common observable pipeline:

$$\begin{aligned} \text{representation or finite species} &\longrightarrow \text{sectorization} \longrightarrow \text{observable family} \\ &\longrightarrow \text{observable shadows} \longrightarrow \text{observable diagnostics.} \end{aligned}$$

This paper treats that pipeline as the theorem-level object. The registry is not bookkeeping, but it is also not the theorem itself. It is the evidence architecture that records which species

realize the pipeline, which observable diagnostics exist, which deformation geometries generate them, and which cross-species claims survive testing.

The registry is open. A new species enters SOF only after satisfying the realization criteria and exposing a well-defined observable ladder.

2 The Universal Observable Pipeline

2.1 Universal Observable Pipeline Principle

The common structure behind the registered examples is not a shared native coordinate system. It is a shared pipeline:

$$\begin{aligned} \text{species} &\longrightarrow V \longrightarrow \{Q_i\} \longrightarrow \mathcal{X} \\ &\longrightarrow \text{observable shadows} \longrightarrow \text{observable diagnostics.} \end{aligned}$$

This is a realization theorem, not a classification theorem. It asserts that an admitted species with the required finite data enters a common observable architecture; it does not classify all possible species or prove that all resulting dynamics agree.

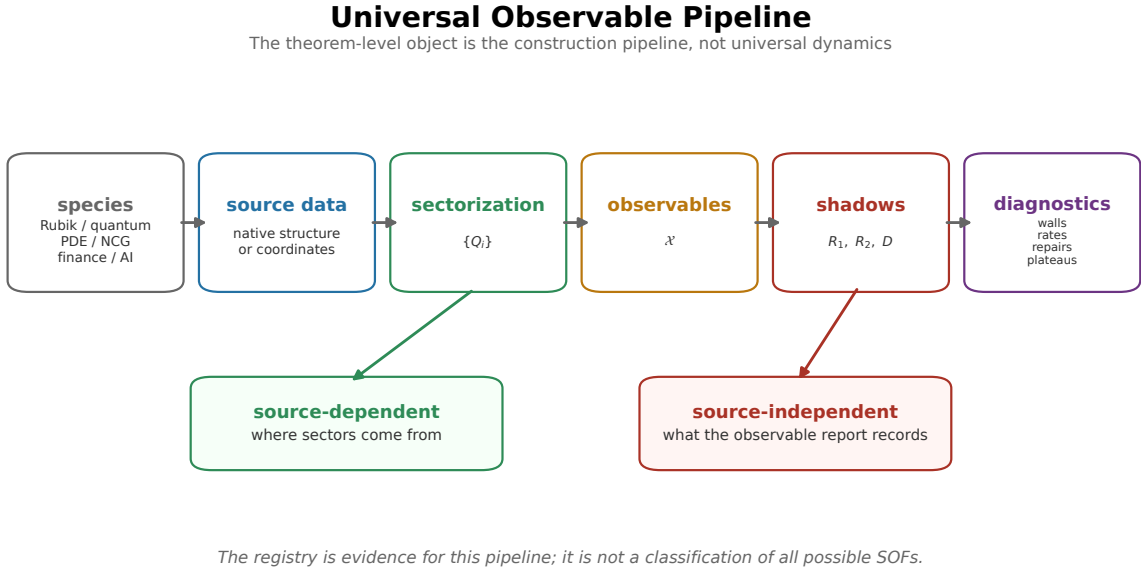


Figure 1. Universal observable pipeline. Paper X treats pipeline-level universality as the theorem-level object: species enter through source data and compatible sectorization, while diagnostics are recorded in a common observable architecture. Dynamics remain species-dependent.

Theorem 2.1 (Universal Observable Pipeline Principle).

Let $\mathcal{S}$ be an admitted finite species equipped with:

1. *a finite-dimensional space V ;*
2. *a finite sectorization $\{Q_i\}$;*
3. *a finite observable extraction rule producing $\mathcal{X}$.*

Then $\mathcal{S}$ induces a Sectorized Observable Framework

$$\mathcal{F}_{\mathcal{S}} = (V, \{Q_i\}, \mathcal{X}),$$

canonical relative to those choices. Consequently every static observable shadow formed from the projectors, the observable family, and finite algebraic operations on them is an SOF-intrinsic construction after realization. Relative diagnostics become intrinsic only after their additional data are specified. In particular, whenever the relevant filtration, deformation, threshold, or diagnostic rule is specified, the shadows and diagnostics

$$R_1, \quad R_2, \quad D, \quad \mathcal{J}_{acc}, \quad \tau(O), \quad \text{wall}(O), \quad \text{repair}(O), \quad \text{plateau}(O)$$

are defined by the same observable syntax, independent of the native species coordinates.

Proof. The sectorization supplies $\{Q_i\}$ and the extraction rule supplies $\mathcal{X}$. Together with V , these are exactly the SOF data. Projected blocks, commutators, depth witnesses, jets, rank/support shadows, wall loci, rate diagnostics, repairs, and plateaus are then obtained by applying the same finite operations to $(V, \{Q_i\}, \mathcal{X})$ and, where needed, to a specified deformation or filtration. The original species determines the realization, but after realization the observable syntax is the same.

This is the precise sense in which Paper X uses “universal”: universality is the existence of a common observable pipeline, not a claim that all registered species obey identical dynamics. $\square$

2.2 Three Claim Layers

Paper X separates three levels of assertion.

Layer	Name	Status	Content
1	Universal Observable Pipeline Principle	theorem-level structure	admitted species pass through representation or finite realization, sectorization, observable family, observable shadows, and diagnostics
2	Calibrated Mechanism-Separation Principle	registry-supported causal principle	distinct driving mechanisms plus ordered response constants can produce distinct proxy rates; supported by Xu-style dynamics, NN diagnostics, and the constructed SOF control
3	Universal Observable Rate Hierarchy	conjectural target	under suitable hypotheses one may obtain $\tau(R_1) < \tau(R_2) < \tau(D)$

The universality developed in this paper is therefore not universality of dynamics. It is universality of the observable construction pipeline. Different species may possess completely different deformation geometries. What is shared is the pipeline

$$\begin{aligned} \text{representation or finite species} &\longrightarrow \text{sectorization} \longrightarrow \text{observable family} \\ &\longrightarrow \text{observable shadows} \longrightarrow \text{diagnostics.} \end{aligned}$$

Observable dynamics remain species-dependent.

There is also a fourth, explicitly programmatic direction: the **Observable Sufficiency Program**. Its question is whether an expanding class of registered species can be compared using the same observable ladder, especially $R_1/R_2/D$ and their dynamic shadows. This is a research program, not a current theorem.

2.3 Why the Pipeline Includes Sectorization

The sectorization step is essential. A finite species may have global observables without sector projectors, but the RIME diagnostics are not global spectral data alone. They are sector-to-sector shadows built from

$$Q_i X_a Q_j.$$

Without $\{Q_i\}$, the registry cannot define support from one sector to another, bridge products through intermediate sectors, frozen pairs, frozen-to-accessible repair, or accessibility walls. With only the trivial sectorization $\{I\}$, these shadows collapse to a one-sector audit: global operators remain, but cross-sector support, bridge, repair, and wall diagnostics disappear.

This is the **Sectorization Necessity Principle**, equivalently the **No-Sector No-Shadow Principle**:

global observables may exist without sectors, but RIME observable shadows require sector projectors.

Paper X uses this as a pipeline boundary. It does not claim that every useful observable theory must be sectorized; it claims that the SOF registry compares species through sector-indexed shadows, and those shadows require sectorization.

2.4 Information Accessibility Interpretation

The common object abstracted by SOF is not a particular algebraic species. It is a layer of **coarse-grained information accessibility**.

A compatible sectorization is a coarse coordinate system recognized by the system under study. In different registry entries it may come from different sources:

Species	Coarse coordinate system
Rubik	QT/HT joint-sector decomposition
Control	reachable-state flag increments
PDE	mesh subdomains and interface sectors
Graph/coloring	vertex or color-class partitions
Finite spectral triple	blocks of a Dirac operator
Neural network	activation-induced regions

Once such sectors are fixed, the natural question is no longer only what each sector contains, but how information, influence, or transport moves between sectors. The first shadows forced by that question are precisely:

direct cross-sector support, two-step or relation-level bridge survival, first depth of accessibility, and walls where these shadows change.

Thus SOF studies the propagation geometry above a compatible coarse coordinate system. Representation theory, control theory, PDE discretization, graph theory, noncommutative geometry, and neural-network dynamics provide different sources of sectorization; SOF compares the information-accessibility geometry that appears after such sectorization is supplied.

This is an interpretation of the pipeline, not an additional theorem. The formal requirement remains the explicit data $(V, \{Q_i\}, \mathcal{X})$ and the claim-status label of each diagnostic.

2.5 Definition of the SOF Registry

Definition 2.2 (SOF Registry).

A **SOF Registry** is a finite or open-ended collection

$$\mathcal{R} = \{\mathcal{E}_s\}_{s \in S}$$

of species entries. Each entry $\mathcal{E}_s$ records:
species, SOF object, observable ladder, dynamics, and diagnostics, together with claim-status metadata.

Here the SOF object means the realized triple

$$\mathcal{F}_s = (V_s, \{Q_i^{(s)}\}, \mathcal{X}_s).$$

The registry is therefore a five-layer taxonomy, not merely a diagnostic table. The five layers are named explicitly below. Claim status is metadata attached to each entry, not a sixth layer, and an entry is admissible only when the SOF object and observable ladder are explicit. Native coordinates alone are not enough.

The registry is open: a new species enters only after satisfying these realization criteria and exposing a well-defined observable ladder.

Thus the registry is the evidence architecture of Paper X. The theorem-level object is the common pipeline; the registry records instances, positive controls, negative controls, and boundary cases for that pipeline.

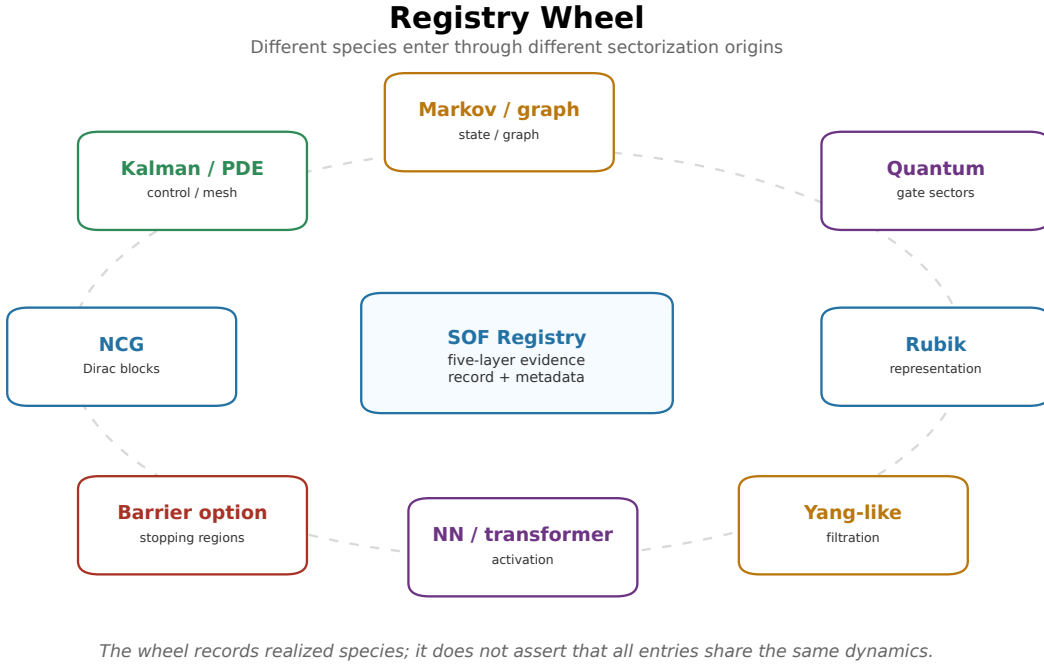


Figure 2. Registry wheel. Registered SOF species enter through different origins: representation, gate action, state partitions, control or mesh geometry, Dirac blocks, stopping regions, activation sectors, and filtrations. The wheel records realized species; it does not assert common dynamics.

2.6 Diagnostic Vocabulary

The shared diagnostic vocabulary is:

Diagnostic	Meaning
$\tau(O)$	characteristic time scale
$\text{wall}(O)$	qualitative-change locus
$\text{repair}(O)$	frozen-to-accessible transition or analogue
$\text{plateau}(O)$	stable observable regime

Not every diagnostic is defined for every species. This is a feature of the registry: absence of a diagnostic is recorded rather than hidden.

2.7 Five-Layer Registry Taxonomy

The registry has five layers:

species $\rightarrow$ SOF object $\rightarrow$ observable ladder $\rightarrow$ dynamics $\rightarrow$ diagnostics.

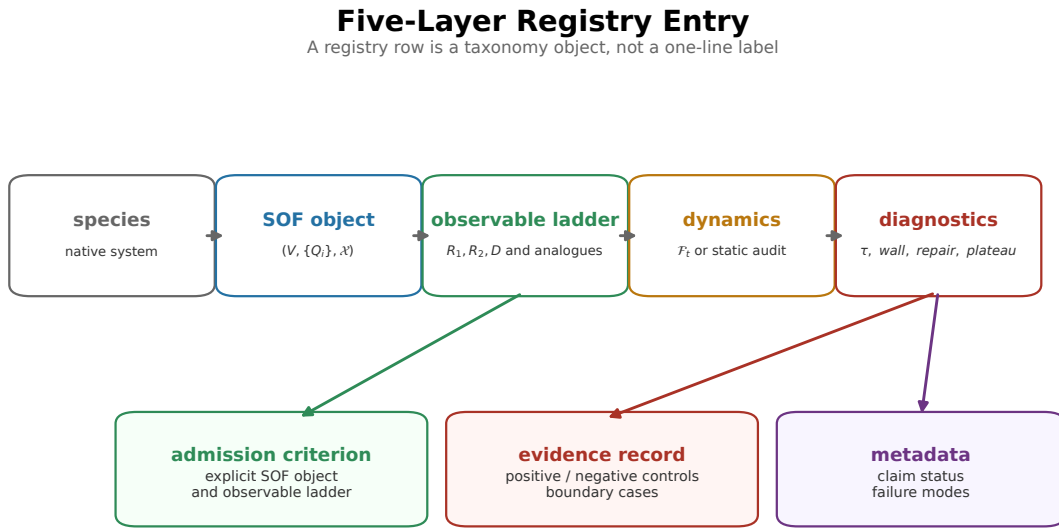


Figure 3. Five-layer registry entry. A registry row records the native species, the realized SOF object, the observable ladder, the dynamics or static audit, the diagnostics, and the claim-status metadata. This prevents registry entries from becoming unsupported labels.

This is the observable layer added above classical structural decomposition. Wedderburn–Artin theory explains how semisimple representations decompose into structural blocks [8, 11, 10]. The SOF registry begins after that stage: it records how sectorized observables, transport ladders, deformation geometries, and diagnostics are built on top of the represented or finite system. The sectorization-origin column records where the coarse coordinate system comes from; it is registry metadata, not an additional SOF axiom.

The complete registry is intentionally larger than the main argument. To keep Paper X centered on the pipeline rather than on a catalogue, the main text uses a compressed registry view. The full five-layer registry is maintained in `docs/SOF_REGISTRY.md`; a reader-facing version can place the full table in an appendix.

Registry role	Representative species	Sectorization origin	Diagnostic status
Core RIME laboratory	Rubik QT/HT	representation / joint spectral geometry	spectra, collision quotient, $R_1/R_2/D$, $\mathcal{J}_{acc}$; near-threshold ratio ≈ 10.8 and plateau oscillation score 0.38
Natural mechanism evidence	Rubik Type III/IV wild mechanisms	representation / joint spectral geometry	288 Type III cancellations and 528 Type IV bridge-level incidence candidates; Type IV remains bridge-level candidate evidence
Mechanism controls	synthetic Type III/IV and mechanism-separated SOF	constructed sector models	boundary controls plus H3 positive control $\tau(K_0^{grow}) = 30 \ll \tau(K_1^{decay}) = 1380$
External rate precedent	Xu ridge model	external row/null decomposition	theorem-proven parameter-space separation; registry audit ratio ≈ 68553
Non-Rubik accessibility	quantum gates	computational-basis or spectral sectors	Pauli $\{X, Z\}$ has $D_{repaired} = 0$, Clifford+CNOT has $D_{repaired} = 6$
Non-representation sectorization	finite spectral triple	geometry / Dirac blocks	$\ [D, p_i]\ _F = 0$, cross-block central distance infinite, two T7-style bridges
Sector-origin portability	control, PDE, combinatorial systems	Kalman flags, mesh/interface partitions, color classes	Kalman ranks 1, 2, 3; PDE left-to-right word-depth 2; coloring has 4 inter-color support edges and 2 same-color conflicts
Stochastic-process portability	barrier-option SOF	stopping/barrier region in a log-price diffusion	cross-barrier support with $R_1 = 75.0\%$, $R_2 = 0.0\%$, and mean first-hit time 6.5915; first hitting is not identified with D
Boundary and contrast species	Markov, graph, NN, Yang-like systems	state, graph, activation, filtration sectors	connected/frozen controls, proxy-only NN rates (60, 80, 120), Yang/RIME plateau contrast 1/5 versus 3/8 zero-crossings

This compressed table is not the theorem. It is the evidence layer showing that the pipeline can be realized in multiple species and that failed or degenerate cases can be recorded without being hidden. Full entries remain governed by the five-layer criterion.

The finite spectral-triple entry plays a distinct logical role. It is conceptually different from the representation-based examples: its sectorization is induced by a block-diagonal Dirac operator rather than by irreducible representation theory or Wedderburn–Artin block decomposition. Nevertheless, the same observable pipeline yields bridge-level accessibility diagnostics. This suggests that the SOF pipeline depends primarily on the existence of a compatible sectorization, rather than on the algebraic origin of that sectorization.

3 Registry Evidence: Universality Probes

3.1 Xu–Vardi–Safran Grokking: Rate-Separation Probe

Xu–Vardi–Safran’s ridge-regression analysis [12] separates a parameter vector into row-space and null-space components:

$$\theta = \theta_{\parallel} + \theta_{\perp}.$$

The row-space component is driven by the empirical loss and evolves quickly; the null-space component is controlled by weight decay and evolves slowly. This is not an SOF theorem. Its role in Paper X is as registry evidence for rate separation.

The SOF bridge is:

parameter-space rate separation $\rightarrow$ observable-space rate separation.

In the completed Paper X registry-evidence audit:

Domain	Fast channel	Slow channel	Ratio	Evidence type
Ridge regression	$\theta_{\parallel}$	$\theta_{\perp}$	68553×	theorem-proven external model; arXiv:2601.19791
RIME near-threshold	R_1	R_2	about 10.8	empirical accessibility diagnostic
NN training-coupled SOF	K_0	K_2	ordered half-response $60 < 80 < 120$	empirical training-coupled diagnostic

The evidence is structural, not numerical equality of ratios. Different domains produce different magnitudes, but both show hierarchical visibility under a specified deformation.

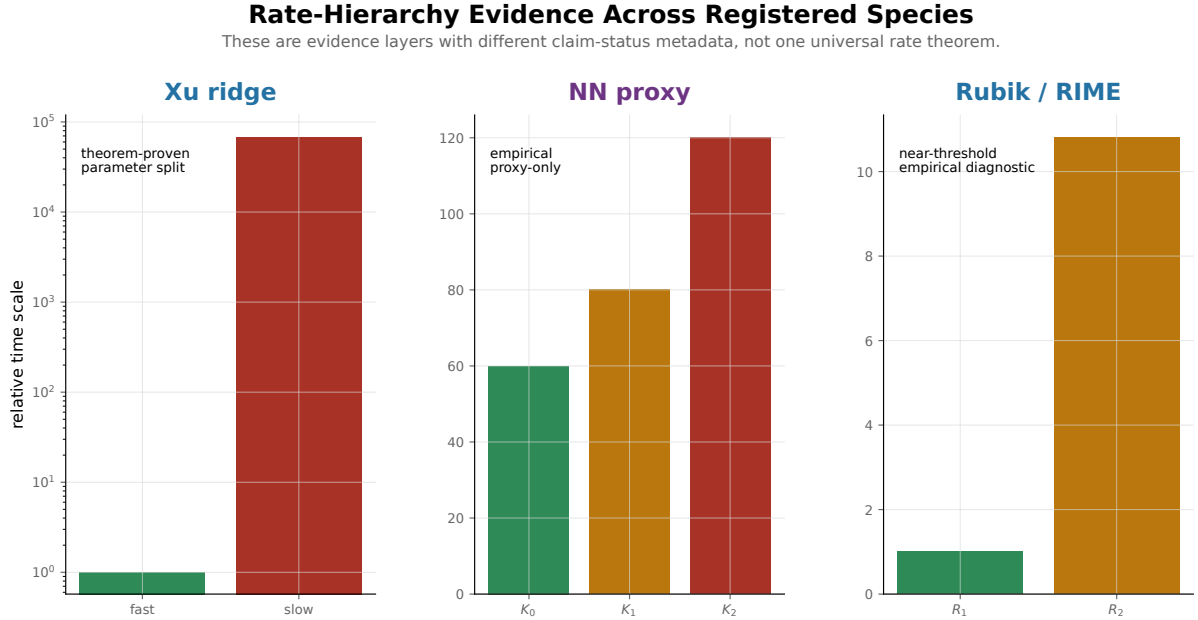


Figure 4. Rate-hierarchy evidence. Xu ridge dynamics, NN proxy diagnostics, and Rubik near-threshold accessibility provide three evidence levels for hierarchical visibility. Their claim status differs: external theorem, empirical proxy, and empirical accessibility diagnostic.

3.2 Yang-Like State Mixing: Plateau-Geometry Probe

Yang-like systems deform state or coherence data:

$$\rho(\varepsilon) = (1 - \varepsilon)\rho_0 + \varepsilon\sigma.$$

This naturally supports monotone filtration degeneration and laws of the form

$$1 - P(\varepsilon) \sim C\varepsilon^\alpha.$$

RIME generator-weight deformation is different. It redistributes observable transport:

$$X_g \mapsto X_g(w).$$

The completed registry-evidence audit gives the following compressed plateau summary:

Deformation	Plateau summary	Zero crossings	Interpretation
Yang-style state mixing	endpoint-flat P_3 sequence	1/5	flat/monotone endpoint degeneration
RIME generator weights	oscillatory P_2 sequence	3/8	oscillatory transport redistribution

Thus state-mixing plateaus may remain flat until an extreme endpoint, while generator-weight plateau data can oscillate. The audited sequences are

$$\begin{aligned} P_3 &= (0.111, 0.111, 0.111, 0.111, 0.111, 0.000), \\ P_2 &= (0.111, 0.111, 0.111, 0.111, 0.111, 0.111, 0.139, 0.111, 0.264), \end{aligned}$$

with the P_2 sequence showing three detrended sign changes out of eight intervals and oscillation score 0.38.

The registry lesson is:

common observable architecture, different deformation geometry, different plateau dynamics.

3.3 Quantum Clifford/CNOT: Non-Rubik Accessibility Probe

Quantum gate systems provide a non-Rubik accessibility probe. In small computational-basis sectorizations, the SOF ladder can be evaluated using gate logarithms or skew generators. The diagnostic observations are:

1. entangling generators such as CNOT reduce frozen pairs and open additional transport channels;
2. the T gate does not enrich the tested low-order R_1/R_2 support beyond the corresponding Clifford+CNOT gate set;
3. the $R_1/R_2/D$ calculus extends to non-Rubik sectorized systems.

In the completed Paper X registry-evidence audit, Pauli $\{X, Z\}$ has $R_1 = 16.7\%$, $R_2 = 33.3\%$, and $D_{\text{repaired}} = 0$, while Clifford+CNOT has the same R_1/R_2 percentages but $D_{\text{repaired}} = 6$ in the same computational-basis sectorization. This is the cleanest current static non-Rubik D-repair witness in the registry.

This is not a claim that quantum systems obey Rubik wall theory. It is a registry claim: quantum systems can expose the same observable ladder under a different deformation and species geometry.

4 Calibrated Mechanism-Separation Principle

The main rate principle is causal before it is empirical:

4.1 Principle 1 (Calibrated Mechanism-Separation Rate Principle)

In an SOF deformation, observable proxy-rate separation is expected when distinct observable channels are driven by distinct mechanisms and their response constants are ordered. For an SOF deformation

$$\mathcal{F}_t = (V_t, \{Q_i(t)\}, \mathcal{X}(t))$$

with accessibility observables defined along the trajectory, the currently supported accessibility pattern appears under the following falsifiable applicability conditions:

Hypothesis	Requirement	Purpose
H1 multi-generator	$ \mathcal{X} \geq 2$	commutator-level observables require multiple generators
H2 cross-sector	some $Q_i X_g Q_j \neq 0$ with $i \neq j$	excludes one-sector or support-null systems
H3 structured dynamics	channels are driven by distinct mechanisms	excludes additive noise that perturbs all Lie layers at once
H3' ordered response calibration	the mechanism constants are ordered in the proposed fast/intermediate/slow direction	prevents mechanism separation from being mistaken for a sufficient condition by itself
H4 non-trivial depth	some pairs have $D \geq 2$ or delayed repair analogue	gives depth repair a nontrivial observable

Under these conditions, the positive registered diagnostics support the following target hierarchy:

Observable layer	Expected scale
direct support	fast
relation survival	intermediate
depth repair	slow

Operationally, H3 requires more than observing different numerical rates after the fact, and it is not sufficient by itself. One must specify a deformation law in which the relevant channels are separated by mechanism and the response constants are calibrated in the proposed order. The paradigm case is Xu–Vardi–Safran ridge dynamics [12]: the data-visible row-space component is gradient-driven, while the hidden null-space component is driven only by weight decay. Paper X now also has a SOF-internal positive control: a three-sector SOF with one gradient-driven growth channel and one regularization-only decay channel. The exact mechanism times are

$$\tau(K_0^{\text{grow}}) = 30, \quad \tau(K_1^{\text{decay}}) = 1380,$$

and the script verifies the same half-response values numerically. This supplies a SOF-internal proxy control for H3/H3'; arbitrary additive noise remains a boundary case because it can perturb direct support, commutators, and nested depth simultaneously.

Proposition 4.1 (Constructed Mechanism-Separated Rate Witness).

Consider a finite SOF with two observable proxy channels $K_a(t)$ and $K_b(t)$. Suppose the first is driven by a fast mechanism with response

$$K_a(t) = 1 - e^{-\gamma t},$$

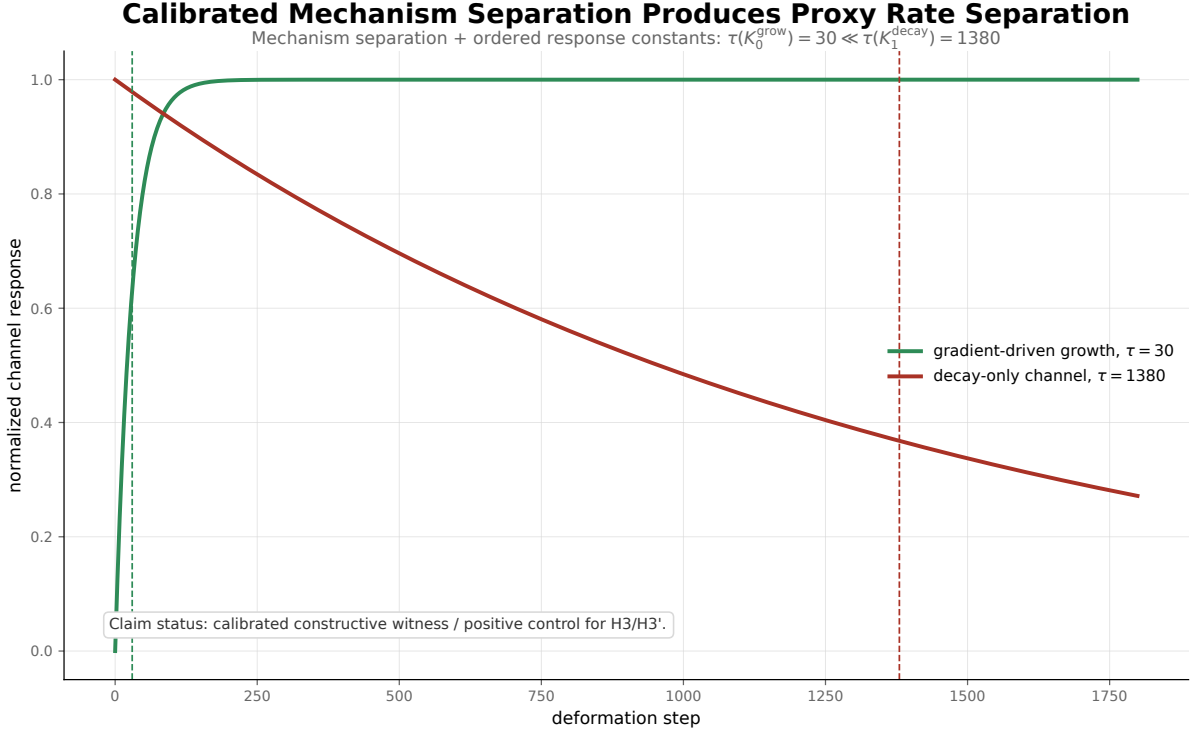


Figure 5. Calibrated mechanism separation. The constructed SOF witness separates a fast gradient-driven growth channel from a slow decay-only channel under ordered response constants, giving $\tau(K_0^{\text{grow}}) = 30 \ll \tau(K_1^{\text{decay}}) = 1380$. This is a proxy-level positive control for H3/H3' structured dynamics.

and the second is driven only by a slow mechanism with relaxation

$$K_b(t) = e^{-\lambda t},$$

where $\gamma > \lambda > 0$. If $\tau_{1/2}$ denotes the half-response or half-decay time, then

$$\tau_{1/2}(K_a) = \frac{\log 2}{\gamma} < \frac{\log 2}{\lambda} = \tau_{1/2}(K_b).$$

Thus calibrated mechanism separation implies observable proxy-rate separation in this constructed SOF model.

Proof. The half-response equation for K_a is $1 - e^{-\gamma t} = 1/2$, hence $t = (\log 2)/\gamma$. The half-decay equation for K_b is $e^{-\lambda t} = 1/2$, hence $t = (\log 2)/\lambda$. Since $\gamma > \lambda$, the first time is smaller. The support script realizes these two channels as sector-to-sector block norms in a three-sector SOF and verifies the calibrated values 30 and 1380.

This proposition is intentionally calibrated and proxy-level. Mechanism separation alone is not asserted to be sufficient; the ordered constants $\gamma > \lambda$ are part of the hypothesis. It does not prove the Observable Proxy Shadow Principle, does not identify K_i with R_i or D , and does not establish a universal rate law for all SOF deformations.

For accessibility SOFs the target form is:

$$\tau(R_1) < \tau(R_2) < \tau(D),$$

or, for raw norm proxies,

$$\tau(K_0) < \tau(K_1) < \tau(K_2).$$

The second display is not a proof of the first. The current registry contains continuous proxy evidence for $K_0/K_1/K_2$ and partial threshold evidence for R_1/R_2 , but it does not yet contain a structured deformation in which $D_{\text{repaired}} > 0$ appears at a measurable time and yields an observed $\tau(D)$.

Equally important, the registry does not yet contain a general bridge from the continuous proxy layer to the discrete shadow layer:

$$K_0, K_1, K_2 \longrightarrow ? \longrightarrow R_1, R_2, D.$$

Paper IX names this missing bridge the Observable Proxy Shadow Principle. It would require threshold calibration, margin stability, sector stability, and compatibility between proxy order and discrete filtration order. Paper X treats this as an open theorem target, not as an assumption of the registry.

This is an intentional separation of roles. Neural-network audits are kept as proxy-rate diagnostics. A genuine $\tau(D)$ audit requires a species where binary repair is already present or naturally expected, such as a structured entangling-gate deformation in the quantum registry or a continuous Rubik generator deformation. $\square$

4.2 Evidence Stratification

Evidence layer	Status	Interpretation
Mechanism-separated SOF control	constructed-witness proposition, $\tau(K_0^{\text{grow}}) = 30 \ll \tau(K_1^{\text{decay}}) = 1380$	H3 positive control inside SOF
Rubik Type III/IV wild mechanisms	288 Type III cancellations and 528 Type IV bridge-level incidence candidates	natural QT/HT-sector mechanism evidence
Training-coupled NN SOF	positive for raw proxies, $60 < 80 < 120$	proxy-only continuous observable rate hierarchy
Engineered near-threshold accessibility	partial, $\tau(R_1) < \tau(R_2)$	direct and commutator channels separate
Finite spectral-triple SOF	central Connes-distance obstruction and 2 ordered T7-style bridges	non-group/Lie portability evidence
Proxy-to-shadow bridge	open	Observable Proxy Shadow Principle not yet proved
Binary D -repair trajectory	open	future quantum or Rubik deformation target; Clifford+CNOT gives static $D_{\text{repaired}} = 6$, but no structured $\tau(D)$ audit yet
Quantum linear interpolation	negative/degenerate	interpolation is not mechanism-separated dynamics
Graph edge rewiring	negative/degenerate	discrete jumps do not define smooth rate hierarchy
Yang-like state mixing	negative/degenerate	state mixing is not training-coupled transport repair
Arbitrary additive noise	negative boundary for $\tau(D)$	simultaneous perturbation can scramble depth
All SOFs	not claimed	rate hierarchy is not a universal property of SOF deformations

The principle is therefore not a blanket theorem. The expanded registry evidence shows that H3 is not cosmetic: without structured dynamics, the observable hierarchy fails, reverses, or becomes degenerate. Each applicability condition has both positive controls and a registered violation in the current registry:

Hypothesis	Positive controls	Registered violation	Failure mode
H1	Clifford+CNOT, NN training-coupled SOF, Rubik generators	one-generator Markov SOF	R_2 collapses because no commutator hierarchy exists
H2	Rubik QT/HT sectors, quantum computational sectors with CNOT	one-sector SOF / degenerate state-sector audit	cross-sector observables collapse
H3	mechanism-separated SOF control, Xu ridge dynamics, NN training with weight decay	arbitrary additive-noise accessibility diagnostic	channels are not mechanism-separated; $\tau(D)$ can be scrambled
H3'	calibrated mechanism-separated SOF control	mechanism separation without ordered constants	proxy rates need not appear in the proposed order
H4	Clifford+CNOT D-repair, Paper V/VII accessibility mechanisms	complete Markov chain / K_3 graph	all pairs are already connected, so no delayed repair event exists

A diagnostic definition of $\tau(O)$ is also required before any rate comparison is meaningful.

5 Claim-Status Boundary

Paper X claims:

1. the Universal Observable Pipeline Principle is the theorem-level object for cross-species comparison;
2. the SOF Registry is evidence architecture for testing that pipeline;
3. the Sectorization Necessity Principle explains why SOF comparison requires projectors before support, bridge, repair, and wall shadows can be formed;
4. registered species can be compared through observable diagnostics rather than native coordinates alone;
5. the completed probes give registry evidence for pipeline portability and cross-species comparability;
6. the Calibrated Mechanism-Separation Principle gives a causal positive control for H3/H3', while cross-species dynamic universality remains registry-level;
7. the Observable Sufficiency Program is a future program, not a theorem.

Paper X does not claim:

1. every mathematical system is naturally an SOF;
2. every SOF has a canonical nontrivial deformation;
3. every useful observable theory must pass through sectorization;
4. all SOFs obey $\tau(R_1) < \tau(R_2) < \tau(D)$;
5. the current registry has already observed a full binary D -repair rate trajectory with $D_{\text{repaired}} > 0$;

6. the continuous proxy ladder $K_0/K_1/K_2$ already determines the discrete shadow ladder $R_1/R_2/D$;
7. Yang, Rubik, finite spectral-triple, quantum, Markov, graph, and neural-network walls are the same geometric object;
8. the registry is closed or complete.

The stable formulation is:

Pipeline is theorem-level structure. Registry is evidence. Calibrated mechanism separation is a registry-supported causal principle. Rate hierarchy is a conjectural target. Observable sufficiency is a program. Sectorization is source-dependent; the observable pipeline is source-independent.

SOF is therefore a sectorization-based observable architecture for cross-species comparison of observable dynamics. This includes R_1 , R_2 , D , trajectories, walls, rates, repairs, and plateaus whenever those diagnostics are defined.

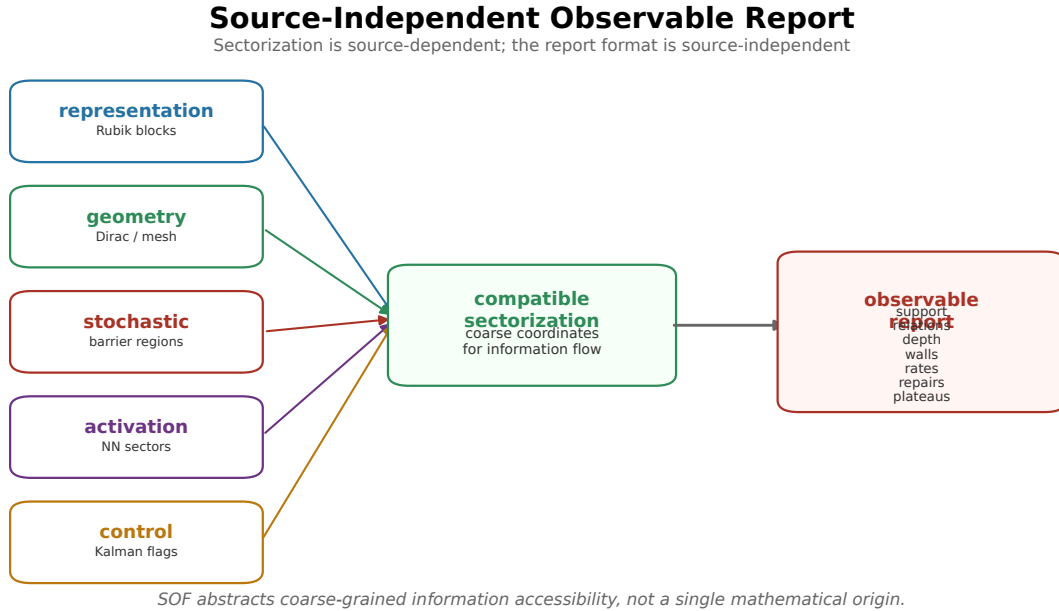


Figure 6. Source-independent observable report. Sectorization may come from representation theory, geometry, stochastic barriers, activation partitions, or control flags. Once compatible sectors are supplied, the observable report has a common format.

6 Outlook

Paper X turns SOF from an object language into a pipeline-based comparison program. The completed registry-evidence probes are:

1. mechanism-separated SOF control: $\tau(K_0^{\text{grow}}) = 30 \ll \tau(K_1^{\text{decay}}) = 1380$;
2. Xu-style rate separation: ridge $68553\times$ versus RIME $10.8\times$, with NN raw proxy rates satisfying $60 < 80 < 120$;

3. Yang-style state mixing versus RIME generator weights: $1/5$ versus $3/8$ plateau zero-crossings, with prethermalization as a separate long-plateau precedent [1];
4. static quantum accessibility repair: Clifford+CNOT has $D_{\text{repaired}} = 6$ versus Pauli $\{X, Z\}$ with $D_{\text{repaired}} = 0$;
5. finite spectral-triple portability: block-diagonal Dirac sectorization gives central Connes-distance obstruction while two ordered T7-style bridge shadows survive;
6. control/PDE/combinatorial portability: Kalman flags, mesh subdomains, and graph-color sectors all support the same observable-pipeline audit;
7. stochastic-process portability: a barrier-option log-price grid gives below/above-barrier sectors, cross-barrier support, and a first-hitting-time diagnostic in a financial-mathematics setting independent of the earlier representation, graph, quantum, and control examples;
8. tau-boundary probes: quantum linear interpolation, graph rewiring, and Yang-like state mixing fail or degenerate, showing that H3 structured dynamics is decisive.

At the present stage, the registry tasks have been discharged at the evidence level: the admitted species now record their sectorization origin, SOF object, observable ladder, deformation geometry when present, diagnostics, and claim status. The registry remains open, but future entries are admitted only when these five layers are explicit. The remaining formal work is to separate strict SOF realization from weaker observable comparison, and to promote the registry-level causal principle to theorem statements only in settings where both a mechanism-separated deformation law and a proxy-to-shadow bridge are available.

The decisive question is whether new species continue to enter the registry by passing through the same pipeline and exposing the same observable ladder. If they do, SOF is not merely a Rubik vocabulary. It is a sectorization-based observable architecture for dynamics across species.

The natural successor is a classification problem. Paper X builds the pipeline and the registry evidence; only after these exist can a later Paper XI ask for an Observable Wall Taxonomy: which SOF wall records, local charts, and observable-rate patterns admit classification, and by what local or global models. Arnold/ADE behavior is only a candidate local chart model for sufficiently smooth discriminant-like wall records. Kontsevich–Soibelman wall-crossing remains a background precedent for exact transformation laws across walls, not a theorem model for SOF at the present stage [9].

Bibliographic Notes

Program lineage. Paper X builds on the SOF object language of Paper VIII [7] and the observable-dynamics language of Paper IX [5]. Its internal registry entries reuse the fixed spectral geometry of Paper IV [3], the accessibility calculus of Paper V [2], the deformation and wall hierarchy of Paper VI [6], and the generic completion and incidence boundary of Paper VII [4].

External background. The representation-theoretic structural layer is the standard finite-dimensional semisimple/Wedderburn–Artin background [8, 11, 10]; SOF adds an observable layer above that decomposition. The rate-separation precedent is Xu–Vardi–Safran’s ridge-regression grokking model [12]. The long-plateau precedent is rigorous many-body prethermalization [1]. The wall-crossing precedent is Kontsevich–Soibelman’s exact transformation-law framework [9]; Paper X uses it only as a discussion contrast.

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